

API Secured, the Modern Way

OAuth2, OIDC, Rate limiting and more



Feedback

Hello!

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Agenda to protect your API

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Authorization with OAuth2

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-  **Authentication with OpenID Connect (OIDC)**

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 **Authorization with OAuth2**

 **Authentication with OpenID Connect (OIDC)**

 **Rate Limiting**

Each topic is explored using **Symfony solution** and a **third-party solution**.

With a surprise at the end! 

What Does "Secured" Mean?

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Authorization OAuth2

What can you do?

Roles, permissions,
access control

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Authentication OIDC

Who are you?

user/password → tokens
client_id/secret → tokens

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Rate Limiting

How much can you do?

Protect resources
Fair usage

Authorization

What is OAuth2?

OAuth2 is an **authorization framework** that allows **users to grant permissions to applications** to access protected resources **without disclosing credentials**.

What is OAuth2?



Today, focus on "on-behalf of user" scenario,
but OAuth2 also supports "client credentials" that is not involving user.

Refers to specification RFC 6749 to learn more.

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● As **CloudPics API developer**, must implement OAuth2 to allow Alice granting access to photos for PhotoPrint.

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OAuth2 use case

In this scenario

☁️ *CloudPics* is the **resource server** that holds Alice's pics and the **authorization server** that issues tokens.

🖼️ *PhotoPrint* is the **client application** that wants to access a protected resource

👩 *Alice* is the **resource owner** of the photos.

OAuth2 implementation in Symfony

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 **Result:** Embed a modern OAuth2 authorization server into your Symfony API

 **In action:**

Your API is the **resource server**

The bundle handles token validation and scope checking for protected endpoints

Bundle installation

Bundle installation

```
composer require league/oauth2-server-bundle
```

Minimal configuration

```
# config/packages/league_oauth2_server.yaml
league_oauth2_server:
  authorization_server:
    private_key: '%kernel.project_dir%/var/oauth/private.key'
    encryption_key: 'def00000examplekey1234567890ab'

  resource_server:
    public_key: '%kernel.project_dir%/var/oauth/public.key'
```

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# config/packages/league_oauth2_server.yaml
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  resource_server:
    public_key: '%kernel.project_dir%/var/oauth/public.key'
```

```
# config/packages/security.yaml
security:
  firewalls:
    api_token:
      pattern: ^/token$
      security: false
  main:
    pattern: ^/api
    security: true
    stateless: true
    oauth2: true
```

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    pattern: ^/api
    security: true
    stateless: true
    oauth2: true
```

```
# config/routes.yaml
oauth2:
  resource: '@LeagueOAuth2ServerBundle/config/routes.php'
  type: php
```

Client entity

```
use League\OAuth2\Server\Entities;

#[ORM\Entity]
class Client implements ClientEntityInterface
{
    #[ORM\Id, ORM\GeneratedValue, ORM\Column]
    private ?int $id = null;

    #[ORM\Column(length: 80)]
    private string $identifier;

    #[ORM\Column(length: 80)]
    private string $secret;

    public function getIdentifier(): string
    {
        return $this->identifier;
    }
}
```

Access token entity

```
use League\OAuth2\Server\Entities;

#[ORM\Entity]
class AccessToken implements AccessTokenEntityInterface
{
    #[ORM\Id, ORM\GeneratedValue, ORM\Column]
    private ?int $id = null;

    #[ORM\ManyToOne(targetEntity: Client::class)]
    private ClientEntityInterface $client;

    #[ORM\Column(type: 'json')]
    private array $scopes = [];

    public function getClient(): ClientEntityInterface
    {
        return $this->client;
    }
}
```

Protected API endpoint

```
// src/Controller/ApiController.php
class ApiController extends AbstractController
{
    #[Route('/api/protected')]
    #[IsGranted('ROLE_OAUTH2_<scope>')]
    public function protectedEndpoint(): JsonResponse
    {
        return $this->json(['data' => 'Protected resource']);
    }
}
```


OAuth2 vs OIDC

OpenID Connect standardizes **user authentication on top of OAuth2**.

It harmonizes the way to **retrieve user related resources** (i.e. profile, email, etc.)

OIDC

You can do **authentication** using **OAuth2**,
but user data are returned in a **non-standard** way.

OIDC standardizes this with the concept of **ID tokens**
that provide user information in a consistent format.

Authentication

OIDC extends OAuth2 with identity layer

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OIDC extends OAuth2 with identity layer

 **ID Tokens:** JWTs containing user identity claims

 **Standard Claims:** sub, aud, iss, exp, etc.

 **Discovery:** Well-known configuration endpoint

 **UserInfo:** Additional user data endpoint

OIDC Provider

Unlike OAuth2 server bundle that integrates an authorization server into the resource server, OIDC recommends to rely on a separated **OIDC Provider** that issues ID tokens, user info and access tokens.

JW* Explanation

Acronym	Full Name	Purpose	Metaphor	Is it a Token?
JWT	JSON Web Token	Defines the structure of the claims in the payload.	The letter or package contents	Yes, although it's never used without a JWS/JWE structure.
JWS	JSON Web Signature	Provides integrity and authenticity via a signature.	A clear envelope with a tamper-proof seal	Yes, a signed token.
JWE	JSON Web Encryption	Provides confidentiality via encryption.	A locked, opaque metal box	Yes, an encrypted token.

JW* Explanation

Acronym	Full Name	Purpose	Metaphor	Is it a Token?
JWA	JSON Web Algorithms	Defines the allowed cryptographic algorithms.	The list of approved lock/seal types	No, it's a list of names.
JWK(S)	JSON Web Key (Set)	A standard format for representing (set of) keys.	The actual key (set) and its metadata	No, it's a (set of) key(s) format.

**Unlike with OAuth2 server bundle,
we are not implementing our own OIDC Provider.**

Open source OIDC Providers

Could also act as Identity Providers (IdP)



Authelia



Casdoor



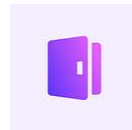
Authentik



Hanko



Keycloak



LogTo



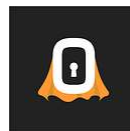
Ory Hydra



PocketID



Rauthy



SuperTokens



Zitadel

They all speak the same language: OIDC protocol

SaaS OpenID Connect Providers (OP)

Could also act as Identity Providers (IdP)



Auth0



AWS Cognito



Bare ID



Cidaas



Cloud-IAM



Microsoft Entra ID
(formerly Azure AD)



Firebase Auth



Hanko



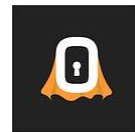
LoginRadius



Okta



Quasr



SuperTokens



Zitadel

They all speak the same language: OIDC protocol

Need social SSO?

Most of listed OIDC providers support social SSO out of the box, by configuration of client credentials.

Symfony supports (most of) OIDC

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 Your API is still the **resource server**

 Your OIDC Provider is the **authorization server**

 Users are **stored in the OIDC Provider**

 Your API **relies on the OIDC Provider** during the security process to **verify user identity and retrieve user information**

Symfony supports (most of) OIDC

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 Using ``OidcUserInfoTokenHandler`` to **validates tokens remotely** with the OIDC Provider

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Symfony supports 2 OIDC strategies:

 Using ``OidcUserInfoTokenHandler`` to **validates tokens remotely** with the OIDC Provider

 Using ``OidcTokenHandler`` to **validates tokens locally** with the OIDC Provider's public key

Both require HttpClient and Cache components.

```
composer require symfony/http-client symfony/cache
```


OidcUserInfoTokenHandler

```
# config/packages/security.yaml
security:
  firewalls:
    main:
      # ...
      access_token:
        token_handler:
          oidc_user_info: https://your-oidc-provider/realms/demo/protocol/openid-connect/userinfo
```

Symfony calls the user info endpoint to validate the token and retrieve user info during authentication.

With OIDC Discovery

```
# config/packages/security.yaml
security:
  firewalls:
    main:
      # ...
      access_token:
        token_handler:
          oidc_user_info:
            base_uri: https://your-oidc-provider/realms/demo/
            discovery:
              cache:
                id: cache.app
```

Symfony automatically discovers the OIDC endpoints (incl. user info) and cache it for better performance.

OidcTokenHandler

Require a library to manipulate JWT tokens:

```
composer require web-token/jwt-library
```

```
# config/packages/security.yaml
security:
    firewalls:
        main:
            # ...
            access_token:
                token_handler:
                    oidc:
                        algorithms: ['ES256', 'RS256'] # Algorithms used to sign the JWS
                        audience: 'api-example' # Audience (`aud` claim): required for validation purpose
                        issuers: ['https://oidc.example.com'] # Issuers (`iss` claim): required for validation purpose
                        keyset: '{"keys":[{"kty":"...", "k":"..."}]}' # A JSON-encoded JWK
```

Symfony validates the token locally with the provided JWKs, without relying on the OIDC Provider.

With OIDC Discovery

```
# config/packages/security.yaml
security:
  firewalls:
    main:
      # ...
      access_token:
        token_handler:
          oidc:
            algorithms: ['ES256', 'RS256']
            audience: 'api-example'
            issuers: ['https://oidc.example.com']
            discovery:
              base_uri: https://www.example.com/realms/demo/
              cache:
                id: cache.app
```

Symfony automatically discovers the OIDC JWKs and cache them for better performance.

Token introspection?

Rate Limiting

Symfony Rate Limiter Overview

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● The Symfony Rate Limiter component provides a token bucket algorithm implementation for rate limiting requests.

Configuring Rate Limiters

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```
# config/packages/rate_limiter.yaml
framework:
  rate_limiter:
    api:
      policy: 'token_bucket'
      limit: 100
      rate: { interval: '1 second', amount: 10 }
```

Using Rate Limiters in Controllers

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 Apply to specific controller actions

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-  Automatic HTTP 429 response when limit exceeded

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Using Rate Limiters in Controllers

-  Apply to specific controller actions
-  Automatic HTTP 429 response when limit exceeded
-  Customize response behavior
-  Access limiter state and headers

Using Rate Limiters in Controllers

```
class ApiController
{
    public function __construct(
        private RateLimiterFactory $apiLimiter,
    ) {}

    public function endpoint(Request $request): JsonResponse
    {
        if (!$this->apiLimiter->create($request->getClientIp())->consume()->isAccepted()) {
            throw new TooManyRequestsHttpException(); // with Retry-After header
        }

        // ... handle endpoint logic
        return new JsonResponse($data);;
    }
}
```




API Gateway, the surprise!

Are last topics your API responsibility?

Authentication, authorization, rate limiting and globally securing and protecting your APIs is a full-time job!

API Gateways or API Management platforms are here to help you with that!

Pick your favorite!

Open-source or commercial, self-hosted or SaaS, there is a large choice!



Apisix



Gravitee



Kong



KrakenD



Otoroshi



Tyk



Zuplo

Delegate!

Delegate!

 Authentication / authorization

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 Authentication / authorization

 Rate limiting

Delegate!

 Authentication / authorization

 Rate limiting

 Logging

Delegate!

 Authentication / authorization

 Rate limiting

 Logging

 Monitoring

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... and more!

Delegate!

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Most of API Gateways offers those basic feature out of the box and finely configurable.

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Not an afterthought. Not a "nice to have."
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Resources

Specifications

[RFC 6749 – OAuth 2.0](#)

[RFC 7519 – JWT](#)

[OpenID Connect – OIDC Spec](#)

[RS256 vs HS256](#)

Documentation

[Symfony Security](#)

[OAuth2 Server Bundle](#)

Tools

[jwt.io – JWT Debugger](#)

[OAuth 2.0 Playground](#)

 **Thank you!**

Questions?



Feedback